

# 10-Week Project Workplan

Project name: 2D Physics-Based Puzzle Game (Matter.js & HTML5 Canvas)

| Week   | Phase                            | Core Tasks & Objectives                                                                                                                                                                                                                                        | Key Deliverable                                     |
|--------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Week 1 | Environment & Architecture Setup | <ul style="list-style-type: none"><li>Initialize web project framework and Canvas viewport.</li><li>Integrate Matter.js engine, runner, and world contexts.</li><li>Set up central Game/Physics loop architecture.</li></ul>                                   | Functional game loop & empty canvas template.       |
| Week 2 | Environment & Static Obstacles   | <ul style="list-style-type: none"><li>Implement static ground, boundaries, and rigid walls.</li><li>Configure Matter.js physics properties (friction, restitution).</li><li>Establish basic scale mapping between engine and viewport.</li></ul>               | Static environment with rigid boundaries.           |
| Week 3 | Dynamic Structure Building       | <ul style="list-style-type: none"><li>Implement dynamic stackable structures (rectangles, circles).</li><li>Tune density, weight, and friction properties for stable structural balancing.</li><li>Develop debug renderer views to verify alignment.</li></ul> | Destructible/stackable block structures.            |
| Week 4 | Sling Mechanics & Controls       | <ul style="list-style-type: none"><li>Implement Mouse Constraint controls for interactive handling.</li><li>Develop drag-and-release tracking with vector constraint calculations.</li><li>Bind projectile ball to slingshot anchor position.</li></ul>        | Interactive drag-and-release slingshot.             |
| Week 5 | Projectile Launch & Forces       | <ul style="list-style-type: none"><li>Code snap-back elastic release logic.</li><li>Apply corresponding physical impulse/force vector to projectile body on release.</li><li>Implement multi-shot tracking sequence.</li></ul>                                 | Fully launchable projectile with realistic impulse. |

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|---------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Week 6  | Target Blocks & Logic     | <ul style="list-style-type: none"> <li>Define specific target blocks with separate collision filters.</li> <li>Track positions, velocities, and tracking height coordinates.</li> <li>Code off-screen detection and out-of-bounds removal.</li> </ul>                   | Target tracking & destruction metrics.          |
| Week 7  | Win/Loss & Level State    | <ul style="list-style-type: none"> <li>Implement win criteria (all targets fallen below altitude threshold).</li> <li>Implement loss conditions (out of remaining projectiles).</li> <li>Design "Level Complete" overlay and "Retry" interface.</li> </ul>              | State system with dynamic game menu overlays.   |
| Week 8  | JSON Level Design         | <ul style="list-style-type: none"> <li>Define structured JSON schemas for object layout coordinates.</li> <li>Hardcode 3 to 5 unique maps with progressive difficulty curves.</li> <li>Build level parser to clear and construct configurations dynamically.</li> </ul> | JSON-driven multi-level sequence manager.       |
| Week 9  | Local Save & Polish       | <ul style="list-style-type: none"> <li>Integrate LocalStorage API to persist unlocked level progression.</li> <li>Refine graphic assets, particle dust trails, and block fragmentation.</li> <li>Optimize rendering pipeline to ensure stable, high FPS.</li> </ul>     | Polished game loop with local progression data. |
| Week 10 | QA, Bug Fixing & Delivery | <ul style="list-style-type: none"> <li>Conduct precise collision testing and extreme-angle stability fixes.</li> <li>Tweak balance parameters (slingshot elasticity, mass ratios).</li> <li>Perform final bundle building and asset optimization.</li> </ul>            | Production-ready web game package.              |